Akash Pandey

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Summary

Final-year B.Tech. student in Computer Science (AI) with hands-on experience building production-grade generative AI systems, ML pipelines, and BI/analytics solutions — including Tensor Veil (a CTGAN-based synthetic data engine), a SQL + Power BI analytics project, and a freight rate prediction ML pipeline. Completed the DeepLearning.AI Machine Learning Specialization and NPTEL Data Analytics with Python (Elite + Silver, 79%), with Oracle OCI AI Foundation Associate and Google Cloud AI certifications. Seeking Machine Learning Engineer, Data Scientist, or Data Analyst roles / internships.

Education

KCC Institute of Technology and Management **Sept 2023 – 2027**
B.Tech. in Computer Science (AI) — CGPA: 7.1/10 *New Delhi, India*

Projects

TensorVeil – Synthetic Data Engine **[GitHub] [Live Demo]**

- Built a generative AI system using CTGAN (GAN-based deep learning) to synthesize privacy-safe training data while preserving full statistical fidelity of source datasets.
- Engineered a scalable, production-grade ML pipeline for data ingestion, automated cleaning, missing-value imputation, and hyperparameter optimization — built to run in production rather than as notebook-level scripts.
- Implemented a validation layer (KS test, Total Variation Distance, ML efficacy via train-synthetic-test-real) comparing real vs. synthetic distributions, functioning as an auditable QA system for generated data.
- Set up a Supabase database with row-level security to track experiments, and deployed as a full-stack Streamlit web application with CI/CD via GitHub Actions and unit tests, open-sourced on GitHub.

Freight Rate Prediction – ML Pipeline **[GitHub]**

- Built an end-to-end machine learning pipeline to predict freight `posted_rate`, training on a 48,000-row dataset and generating predictions for a 12,000-load validation set.
- Designed and validated a train/validation split methodology, documented in a full technical report alongside model results and approach.
- Produced a time-series rate chart for a fixed December lane (Lexington to Fort Wayne) to visualize predicted vs. actual rate trends, delivered alongside a code repository, prediction outputs, written report, and a recorded walkthrough.

Olist E-Commerce Sales & Delivery Analysis – SQL + Power BI Dashboard **[GitHub]**

- Built an end-to-end analytics project on ~99,000 orders from the Olist Brazilian e-commerce dataset, using PostgreSQL for data storage/analysis and Power BI for dashboarding and DAX.
- Modeled 7 relational tables (orders, customers, payments, reviews, products, sellers, geolocation, etc.) into a star-schema-style structure in PostgreSQL, including a custom Date dimension table.
- Wrote SQL to answer core business questions on sales performance, customer retention, delivery reliability, and payment behavior, then designed a 4-page interactive Power BI dashboard (Sales Overview, Customer Insights, Delivery & Reviews, Payments) with KPIs, geo maps, and DAX measures.
- Surfaced quantified business insights — e.g., late deliveries drop average review score by nearly 2 points (4.3 → 2.57), and only ~3% of customers make a repeat purchase, pointing to a clear retention gap.

Top Rated TMDB Movies – Cleaned Dataset **[Kaggle]**

- Performed EDA on raw TMDB movie data, cleaned and preprocessed it (missing values, duplicates, field standardization), and published the cleaned dataset publicly on Kaggle for other practitioners.

Technical Skills

Languages	Python (primary), C++, Java
ML/DL Frameworks	Scikit-Learn, TensorFlow, CTGAN
Data Engineering & Analysis	NumPy, Pandas, Data Cleaning, EDA, Feature Engineering
Statistics & Validation	Model Evaluation
Databases	PostgreSQL, Supabase (PostgreSQL-based)
BI & Visualization	Power BI (DAX, dashboarding), Streamlit
Generative AI & Automation	Synthetic Data Generation, Prompt Engineering, Claude Code
Tools & Platforms	Google Cloud Platform (GCP), Git/GitHub
Concepts	Supervised & Unsupervised Learning, Statistical Modeling, DSA

Certifications & Achievements

- Machine Learning Specialization – DeepLearning.AI / Andrew Ng (Coursera)
- Data Analytics with Python – NPTEL (Elite + Silver, Score: 79%)
- Oracle OCI AI Foundation Associate (2025, Score: 85%)
- AI-Deate AI Case Competition by B&C of IIT Delhi – Rank #67; HackerRank – Introduction to Programming Using Python, Certificate of Achievement